

Revision Note

Paper: Receiver-Domain Behavioral Probing for Backdoor-Resilient Federated GPS Spoofing Detection in UAV Networks

Group 1: Will Jedrzejczak, Cole Walther, Dilpreet Gill

This was a rebuild around one story, not a compression. Nothing was shrunk to fit; sections were removed, merged or rewritten based on whether they advance the argument a reviewer has to be convinced by.

Result: 15 pages, 12 tables/figures, 25 subsections becomes ~7 pages, 1 table and 4 figures, 15 references. The title changed to lead with the mechanism rather than the property, since the mechanism is the contribution; "in UAV Networks" was kept, since UAV is a topical keyword that affects how a communications symposium routes the paper.

The story the paper now tells

Self-reported performance is an exploitable attack lever in federated UAV spoofing detection. We propose receiver-domain behavioral probing, which lets the coordinator judge client models on evidence it generates itself. Four result elements carry the whole argument, in order:

attack validity -> competitive comparison -> key differentiator -> limitation and generalization

Everything that did not serve that chain was cut.

Retained, and why

Element	Why it stays
Table I (10-row comparison, full width)	Proves the attack works, inflation makes it worse, median and FLTrust leave residual lift, Multi-Krum is competitive, and ours attributes per client
Fig. 2 (unknown attacker count)	The strongest differentiator: existing rules need f, we do not
Fig. 3 (heterogeneity)	The honest boundary, and the empirical justification for keeping the median backstop
Fig. 4 (trigger generalization)	Supports the scoped claim that the trigger feature need not be known
Fig. 1 (system/threat model)	The authors' own diagram, showing the three attack steps and the inflated weight path
Algorithm 1	Shortened from 20 lines to 11; the method is the technical center

Removed from the main paper

All of this remains in the repository artifact and is summarized in one or two sentences where it matters.

- **Positioning table against related work** -- replaced by two compact paragraphs, per instruction
- **Layer-ablation table** -- subsumed by Table I, which contains the same conditions
- **Hyperparameter sweep** table and figure -- two sentences in §V-G
- **Trust-degradation table** -- one clause; its decisive case is now Fig. 3
- **Centralized detector-ceiling table** -- one paragraph in the Discussion limitations, keeping the five numbers that matter (0.907 centralised vs 0.529 federated, 0.974 deeper, 0.993 boosted, 0.224 logistic) and the argument that lift is unaffected by where the baseline sits
- **Cost scaling plots** and the two-denominator table -- two sentences in §V-G
- **Adaptive-attacker table and figure** -- the sweep is reported in prose in §V-E, with all eight cited values checked against the CSV
- **Feature-separability table** (10 rows) -- replaced by two sentences giving the range ($d = 0.306$ down to 0.152 probed, 0.018 and 0.001 excluded)
- **False-positive breakdown table** -- folded into the honest-false-flag column of Table I and the per-client rows of Table II
- **Failed Dirichlet experiment** -- cut entirely; the working partition is described in one sentence
- **Preprocessing table** and per-step row counts -- reduced to two sentences
- **Round-progression, sensitivity and FLTrust-only figures** -- the tables carry the same numbers

Rewritten rather than trimmed

- **Abstract** : 340 -> 192 words. Problem, attack, idea, one attack number, one defense number, one differentiator, one scope sentence. Multi-Krum stays because it frames the differentiator; the adaptive attacker, cost, trigger sweep and false-positive rate are all out.
- **Introduction** : ~1,900 -> 617 words. Related work compressed from four paragraphs of per-paper summary into one gap paragraph and one FLTrust-positioning paragraph. Contributions cut from five bullets to three.
- **System and threat model** : the three attack steps are now labeled A1-A3 and expressed with symbols, with one equation each where it earns one.
- **Results** : reorganised so each subsection maps to exactly one element. §V-A to §V-D carry the four elements above; §V-E to §V-G restore the adaptive attacker, per-client attribution and root-set stress as full subsections rather than the single compressed paragraph they had been reduced to.
- **Discussion and Conclusion** : merged into one section.

Presentation pass

- Every table and figure label is in Title Case, and the cryptic column heads of the first draft ("Recall / BSR / Lift / Detect / ms/rd") became two-line headers that name the quantity ("Spoofing Recall", "Backdoor Lift", "Attacker Detection", "Server (ms/Round)").
- Table I moved to a full-width table*. It now carries clean accuracy, the standard deviation on lift, the honest false-flag rate, and the two rules that had been relegated to a sentence (Krum and trimmed mean), so nothing in it is "omitted for space" any more.
- The figures were rebuilt from scratch rather than patched; see **Figures** below.
- Every value in Tables I and II, and every adaptive-attacker value cited in prose, is checked against its exported CSV by a script, so a hand-typed number cannot drift from the run that produced it.

Where we departed from the brief, and why

One departure remains, flagged here rather than left to be noticed. The brief suggests 1.3-1.6 pages for setup plus results plus discussion; after the Sunday-review compression we are at 2.89. The four core result elements plus Table I account for 1.45 pages of that on their own, so the prose around them is roughly 1.4 pages. Cutting further would mean dropping one of the four elements.

Second advisor review (Sunday meeting)

Four comments, all acted on.

- **Author block reformatted** to match the reference ICC paper: names, one shared affiliation, then emails, with no superscript markers. Dr. Hasan added as an author.
- **System and threat model enlarged** from 0.78 to 1.29 pages. Added the round protocol and what the coordinator can and cannot observe, the motivation for accuracy weighting so the attacked design is treated fairly, a formal capability and knowledge model for the adversary, its objective as a constrained maximization, and the reasoning behind each of A1, A2 and A3 (including that $\gamma = 3$ is well below the full-replacement value of N).
- **Methodology enlarged** from 0.81 to 1.58 pages, and given a spine. A new *Design Requirements* subsection states the four constraints R1-R4 that any replacement for Eq. (2) has to satisfy, and the rest of the section is organised as meeting them; R2, R3 and R4 are then referenced by the results that test them. Probe construction now justifies its three design choices individually, the suspicion equation names the alternative each element rejects, and a *Composition, Cost and Scope* subsection covers why two layers, why A3 is structurally inert, and the per-round cost.
- **Results cut** from 4.20 to 2.89 pages. The adaptive-attacker and per-client-attribution tables are gone (both restated their own prose), captions were tightened, and every results subsection was compressed. The four core result elements are untouched.
- **More references for the self-report framing**, so the problem does not rest on one paper: Yin *et al.* for coordinate-wise median and trimmed mean, which we had been using uncited, plus Kang *et al.* and Kairouz *et al.* for the wider reputation- and contribution-weighted family. 12 references becomes 15, inside the 12-15 the brief asks for.

What the first compile changed

Three things were only visible once the paper was actually typeset, and all three are fixed:

- **Fig. 1 was being shrunk.** A height=2.0in cap had been added as a guard against an unknown-aspect diagram. The diagram turned out to be 1.36:1, so the cap bound first and rendered it 2.72 in wide inside a 3.5 in column, leaving it adrift in white space with unreadable labels. The cap is gone; it now fills the column at 3.49 x 2.57 in.
- **The table captions became walls of small caps.** IEEEtran sets table captions in small caps, so the long explanatory captions turned into 4, 8 and 5 line blocks -- Table II's caption was taller than Table II. All three are cut to two lines, and the definition of the detection and false-flag columns moved to Section II-C, where the other metrics are defined and where it belongs anyway.
- **Figs. 2 and 3 were deferred onto one page together.** Table I took the wide-float slot on its page, and both figures landed on the next one stacked back to back. Fig. 2 is now declared immediately after Table I so it can share that page, leaving Fig. 3 its own.

Everything else held up: 7 pages, no overfull boxes, no float page, figures placed at scale 0.997.

References

12, down from 12 -- the count was already in range, but two were re-verified against source during the previous revision and the dataset citation was corrected to the Mendeley *Unmanned Aerial System* release (10.17632/z7dj3yyzt8.3), which matches the data actually used. The bibliography is now inline (thebibliography) rather than BibTeX, so the paper compiles in one pass with no .bbl step.

Figures

Figures 2 to 4 rebuilt at publication quality and exported as **600 dpi PNG**, twice the 300 dpi IEEE asks for line art. The first draft put all three at one column width, which crushed five overlapping series into 3.4 in and stacked Fig. 3's two panels vertically. The rebuild:

- **Figs. 2 and 3 span the full text width.** They carry the two arguments a reviewer has to be convinced by, and they were the two that looked cramped. Fig. 3's panels now sit side by side.
- **Fig. 2 labels its five lines directly at the right** instead of carrying a legend. A legend with five entries is a second lookup for the reader and was eating a third of the plot height. Labels that would collide are pushed apart automatically and keep a leader line to their curve.
- **Fig. 3(b) uses twinned axes**: attacker detection on the left, attacker trust on the right, with the uniform share of 0.100 drawn in. The point of the panel is that one collapses exactly as the other rises, and that is now one picture rather than two numbers in a caption.
- **Fig. 4** marks the region at or below an honest fleet, so "defended lift is negative" is visible rather than something the reader has to work out from the axis.
- Nothing is set below 8 pt, grids are horizontal only and stop before the label gutter, markers carry a white edge so overlapping points stay separable, and legend handles are proxies without the error-bar caps that errorbar() puts in by default.
- **Each PNG is written at exactly the width it is placed at**, so LaTeX never rescales it. This was a real bug: `bbox_inches='tight'` crops the unused margin, so the files came out 6.01, 7.18 and 3.21 in wide and were then stretched to `\textwidth / \columnwidth` by 1.19x, 1.00x and 1.09x. The same 9.4 pt axis label was therefore reaching the page at 11.2, 9.4 and 10.2 pt in the three figures, and Fig. 2's was larger than the 10 pt body text. `save()` now measures the crop and re-saves until the width is exact.
- Table column gutters were opened up (`tabcolsep` 4 to 7 pt on Table I) after measuring each table's natural width in Times at 8 pt: Table I needs 475 pt of the 516 pt available, so the columns had no reason to be tight.
- Consistent colour role across figures: red = attack/undefended, green = proposed, purple = Multi-Krum and (in Fig. 3b) attacker trust, blue = FLTrust, grey = median. No colour means two different things inside one figure.

Figure 1 is the authors' own diagram rather than a generated one. It shows the three attack steps and, importantly, the path by which the inflated self-reported accuracy drives the aggregation weight toward its maximum, which is the specific vulnerability the paper targets.

One thing we did not do

The paper does **not** claim the proposed method dominates every baseline, because our own Table I would contradict it. Multi-Krum is statistically indistinguishable from behavioral trust alone under the nominal IID setting and costs far less server time. That result is stated plainly in the abstract, in §V-A and in the Conclusion, and is then used to motivate the two properties that do distinguish the method: no attacker-count parameter, and per-client attribution.

Deliverables in this folder

File	What it is
`main.tex`	The paper, IEEEtran `conference` class, inline bibliography
`figures/fig2..4_*.png`	Figs. 2-4 at 600 dpi. Figs. 2-3 are full text width, Fig. 4 one column; Fig. 1 is supplied by the authors
`build_figures.py`	Regenerates Figs. 2-4 from the CSVs
`14_conference_results.ipynb`	Executed notebook reproducing Tables I-III and Figs. 2-4
`fl_common.py`, `fl_runner.py`	The shared harness (split, model, attack, 9 aggregation rules, metrics)
`exp_*.py`, `run_all.py`	The experiments behind the reported results
`results/*.csv`	Every reported value, exported

Reproduce with `python run_all.py` baselines attackers noniid then `python build_figures.py` , or set `RUN_EXPERIMENTS = True` in the notebook.